

Chandan Kumar

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EDUCATION

Motilal Nehru National Institute of Technology Allahabad

2021 – 2025

B.Tech. in Computer Science and Engineering

CPI: 8.23

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Distributed Systems, OOP

St. John Senior Secondary School, Dumraon

2018 – 2020

Class XII, CBSE

92.2%

EXPERIENCE

Marvell Technology

May 2024 – Jun 2024 (Intern) | Jan 2025 – Present (Full-Time)

Software Engineer — Storage Accelerator

Pune, India

- Designed and automated scalable Python test frameworks (PyTest + Paramiko) to validate distributed **NVMe storage systems**, reducing manual validation effort by **~90–95%** and enabling automated CI gating of firmware releases.
- Deployed and orchestrated **multiple Docker containers in parallel** to run isolated simulation workloads simultaneously, validating independent storage features across containerised environments on Ubuntu.
- Owned end-to-end test module lifecycle — design, implementation, execution, failure analysis — across **NVMe storage and virtualisation** layers; collaborated cross-functionally with firmware, systems, and cloud teams.
- Integrated test automation into internal **CI/CD pipelines**, supporting continuous delivery and automated regression gating for storage stack releases.
- Performed **root-cause analysis** of storage system failures and regressions, improving platform stability and preventing recurring issues across distributed storage infrastructure.
- Maintained validation tooling on **Linux (Ubuntu)**, including system-level scripting for storage provisioning, process monitoring, and log-based failure triage using Bash and Python.

PROJECTS

Video Summarization | *AI-powered video summarizer*

Nov 2024 | [GitHub](#)

- Built a video summarisation system using **BiLSTM/LSTM with Luong attention** and Particle Swarm Optimization (PSO) to generate concise summaries; trained and evaluated on SumMe and TVSum benchmark datasets.
- Implemented efficient video segmentation pipelines (KTS) and ResNet-based feature extraction with PyTorch and OpenCV.
- **Stack:** Python, PyTorch, LSTM/BiLSTM, OpenCV, ResNet, NumPy, Pandas.

Complaint Box | *Full-stack complaint management platform*

Nov 2022 | [Live](#) | [GitHub](#)

- Designed and built a full-stack platform with **role-based access control (RBAC)**, email verification, and secure session management; designed RESTful APIs and a MongoDB schema for concurrent users.
- **Stack:** Node.js, Express.js, MongoDB, JavaScript, EJS, HTML, CSS.

SKILLS

Languages : Python, C++

Infrastructure : Distributed storage systems, NVMe storage validation, storage test automation, reliability & regression testing

Linux & OS : Ubuntu (primary dev/test env), Bash scripting, system monitoring, log analysis, storage provisioning

Containers : Docker (parallel containerised workloads in production validation), containerisation & virtualisation concepts

Dist. Systems : Replication, consistency models, fault tolerance, NVMe/storage stack internals, cross-layer debugging

Backend : Node.js, Express.js, REST APIs, PyTest, Paramiko (SSH automation)

Databases : SQL, MongoDB

Dev Tools : Git, Gerrit, Jira, Confluence, CI/CD pipelines

ML / AI : PyTorch, LSTM/BiLSTM, OpenCV, ResNet

ACHIEVEMENTS

LeetCode Knight (**Top 5%**, Rating 1871) | CodeChef Starters 88 — Global Rank **468**